APO-FLUCONAZOLE
Fluconazole 50mg and 100mg
Antifungal Agent

CLINICAL PHARMACOLOGY
Fluconazole is a highly selective inhibitor of fungal cytochrome P-450 sterol C-14-α -demethylation. Mammalian cell demethylation is much less sensitive to fluconazole inhibition. The subsequent loss of normal sterols correlates with the accumulation of 14-α -methyl sterols in fungi and may be responsible for the fungistatic activity of fluconazole. Fluconazole is a polar *bis*-triazole antifungal agent which exhibits fungistatic activity *in vitro* against a variety of fungi and yeasts; it also exhibits fungistatic activity *in vivo* against a broad range of systemic and superficial fungal infections. In common with other azole antifungal agents, most fungi show a higher apparent sensitivity to fluconazole *in vivo* than *in vitro*. Both orally and intravenously administered fluconazole were active in a variety of animal fungal infection models. Activity has been demonstrated against opportunistic mycoses, such as infections with *Candida spp.*, including systemic candidiasis and in immonucompromised animals; with *Cryptococcus neoformans*, including intracranial infections; with *Aspergillus spp.*, including systemic infections in immunocompromised animals; with *Microsporum spp.*; and with *Trichophyton spp*. Fluconazole has also been shown to be active in animal models of endemic mycoses, including infections with *Blastomyces dermatitides*; with *Coccidioides immitis*, including intracranial infection; and with *Histoplasma capsulatum* in normal and immunosuppressed animals.

Pharmacokinetics
Absorption: Well absorbed from the GI tract. Bioavailability: ≥90% (oral). Time to peak plasma concentration is within 1-2 hour.
Distribution: Widely distributed into body tissues and fluids. Enters breast milk. Volume of distribution is approx 0.6 L/kg. Plasma protein binding is approx 12%.
Excretion: Via urine (approx 80% as unchanged drug, approx 11% as metabolites). Elimination half-life is approx 30 hour.

INDICATION AND CLINICAL USE
APO-FLUCONAZOLE (fluconazole) is indicated for the treatment of:
1. Oropharyngeal and esophageal candidiasis. APO-FLUCONAZOLE is also effective for the treatment of serious systemic candidal infections, including urinary tract infection, peritonitis, and pneumonia.
2. Cryptococcal meningitis.
3. Prevention of the recurrence of cryptococcal meningitis in patients with acquired immunodeficiency syndrome (AIDS).

APO-FLUCONAZOLE is also indicated to decrease the incidence of candidiasis in patients undergoing bone marrow transplantation who receive cytotoxic chemotherapy and/or radiation therapy.

ADVERSE REACTIONS
The commonest side effects are gastrointestinal disorders include nausea, vomiting, abdominal pain, diarrhea and flatulence. Skin rash and headache are also commonly reported side effects. In some patients, particularly those with serious underlying disease, eg. AIDS and cancer, changes in renal and hematological function test results and hepatic abnormalities (see Warnings) have been observed during treatment with fluconazole, but the clinical significance and relationship to treatment is uncertain.
Patients with AIDS are more common prone to the development of severe cutaneous reactions to many drugs. Exfoliative skin disorders including Stevens-Johnson Syndrome has been reported in AIDS patients receiving fluconazole concomitantly with other agents. If a rash develops in a patient treated for a superficial fungal infection which is considered attributable to fluconazole, further therapy with this agent should be discontinued. If a patient with systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop. In rare cases, anaphylaxis has been reported.

PRECAUTIONS
<u>Use during Pregnancy:</u> There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of fluconazole as a single or repeated dose in the first trimester.
Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infections in whom APO-FLUCONAZOLE may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus.
Effective contraceptive measures should be considered in women of childbearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose.
There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high-dose (400mg/day to 800mg/day) fluconazole therapy for coccidiodomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5mg/kg or 10mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 50 mg/kg and higher doses. At doses ranging from 80 mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg, embryolethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification.
Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800 mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

<u>Use during Lactation</u>: Fluconazole is found in human breast milk at concentration similar to plasma. Breast-feeding may be maintained after a single dose of 150mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole.

Superinfection: Development of resistance to fluconazole has not been studied; however, there have been reports of cases of superinfection with Candida species other than *C.albicans*, which are often inherently not susceptible to fluconazole (e.g. *Candida krusei*). Such cases may require alternative antifungal therapy.
As for other anti-infectives used prophylactically, prudent medical practice dictates that fluconazole be used judiciously in prophylaxis, in view of the theoretical risk of emergence of resistant strains.

WARNINGS
Hepatic Injury: Fluconazole has been associated with rare cases of serious hepatic toxicity, including fatalities primarily in patients with serious underlying medical conditions. In cases of fluconazole associated hepatotoxicity, no obvious relationship to total daily dose, duration of therapy, sex or age of the patient has been observed. Fluconazole hepatotoxicity has usually, but not always been reversible on discontinuation therapy. Patients who develop abnormal liver function tests during Apo-Fluconazole (fluconazole) therapy should be monitored for the development of more severe hepatic injury. Apo-Fluconazole should be discontinued if clinical signs and symptoms consistent with liver disease develop that may be attributable to fluconazole.
Anaphylaxis: In rare cases, anaphylaxis has been reported.
Dermatologic: Patients have rarely developed exfoliative skin disorders during treatment with fluconazole. In patients with serious underlying diseases (predominantly AIDS and malignancy) there have rarely resulted in a fatal outcome. Patients who develop rashes during treatment with Apo-Fluconazole should be monitored closely and the drug discontinued if lesions progress.

CONTRAINDICATIONS
APO-FLUCONAZOLE (fluconazole) is contraindicated in patients who have shown hypersensitivity to fluconazole or to any of its excipients. There is no information regarding cross hypersensitivity between fluconazole and other azole antifungal agents. Caution should be used in prescribing APO-FLUCONAZOLE to patients with hypersensitivity to other azoles.
Co administration of terfenadine is contraindicated in patients receiving APO-FLUCONAZOLE at multiple doses of 400mg or higher based upon results of a multiple dose interaction study.

DRUG INTERACTIONS
Theophylline: Plasma-theophylline concentration is increased by fluconazole. Patients who are receiving high doses of theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and therapy modified appropriately if signs of toxicity develop.
Cimetidine: Absorption of orally administered fluconazole does not appear to be affected by gastric pH.
Antacid: Administration of antacid had no effect on the absorption or elimination of fluconazole.
Cyclosporine: Fluconazole administered at 100 mg daily dose does not affect cyclosporine pharmacokinetic levels in patients with bone marrow transplants. Fluconazole may significantly increase cyclosporine levels in renal transplant patients with or without renal impairment. Careful monitoring of cyclosporine concentrations and serum creatinine is recommended in patients receiving fluconazole and cyclosporine.
Tacrolimus: There have been reports that an interaction exists when fluconazole is administered concomitantly with tacrolimus, leading to increased serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were coadministered. Patients receiving tacrolimus and fluconazole concomitantly should be carefully monitored.
Warfarin: Prothrombin time may be increased in patients receiving concomitant fluconazole and coumarin-type anticoagulants. Careful monitoring of prothrombin time in patients receiving fluconazole and coumarin-type anticoagulants is recommended.
Hydrochlorothiazide: Plasma concentration of fluconazole increased by hydrochlorothiazide.
Oral Contraceptives: Plasma concentration of oral contraceptives (ethinyl estradiol and levonorgestrel) was significantly increased when concomitantly administered with fluconazole.
Oral Hypoglycemics: Clinically significant hypoglycemia may be precipitated by the use of fluconazole with oral hypoglycemic agents. Fluconazole reduces the metabolism of tolbutamide, glyburide, and glipizide and increases the plasma concentration of these agents. When fluconazole is used concomitantly with these or other sulphonylurea oral hypoglycemic agents, blood glucose concentrations should be carefully monitored and the dose of the sulphonylurea should be adjusted as necessary.
Phenytoin: Fluconazole increases the plasma concentrations of phenytoin. Careful monitoring of phenytoin concentrations in patients receiving fluconazole and phenytoin is recommended.
Rifampin: Rifampin enhances the metabolism of concurrently administered fluconazole. Depending on clinical circumstances, consideration should be given to increasing the dose of fluconazole when it is administered with rifampin.
Zidovudine: There was a significant increase in plasma concentration of zidovudine following the administration of fluconazole. The metabolite, GZDV, to parent drug ratio significantly decreased after the administration of fluconazole. Patients receiving this combination should be monitored for the development of zidovudine-related adverse reactions.
Drugs prolonging the QTc interval: The use of fluconazole in patients concurrently taking drugs metabolized by the Cytochrome P450 system may be associated with elevations in the serum levels of these drugs. In the absence of definitive information, caution should be used when co administering fluconazole and such agents. Patients should be carefully monitored.
Terfenadine: The combined use of fluconazole at doses of 400 mg or higher with terfenadine is contraindicated. Patients should be carefully monitored if they are being concurrently prescribed fluconazole at multiple doses lower than 400 mg/day with terfenadine.
Astemizole: Definitive interaction studies with fluconazole have not been conducted. The use of fluconazole may be associated with elevations in serum levels of astemizole.
Cisapride: There have been reports of cardiac events including torsades de pointes in patients to whom fluconazole and cisapride were coadministered. Therefore, caution should be used when co administering fluconazole with cisapride.
Others: Interaction studies with other medications have not been conducted. But such interactions may occur.

RECOMMENDED DOSAGE

Apo-Fluconazole (fluconazole) is well absorbed and excreted predominantly unchanged in urine following oral administration in man. The oral bioavailability is essentially complete (greater than 90%), and is independent of dose. Peak plasma concentrations after oral administration are attained rapidly, usually within 2 hours of dosing. Since oral absorption is rapid and almost complete, the daily dose of fluconazole is the same for oral and intravenous administration. The terminal plasma elimination half-life is approximately 30 hours (range 20-50 hours).
The daily dose of Apo-Fluconazole and the route of administration should be based on the infecting organism, the patient's condition and the response to therapy. Treatment should be continued until clinical parameters and laboratory tests indicate that an active fungal infection has been cured or has subsided. An inadequate period of treatment may lead to recurrence of active infection. Patients with AIDS and cryptococcal meningitis or recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.

Recommended Dosages in Adults and Children

Loading Dose: Adults and Children – Administration of a loading dose on the first day of treatment, consisting of twice the usually daily dose, results in plasma concentrations close to steady state by the second day. Patients with acute infections should be given a loading dose equal to twice the daily dose, not to exceed a maximum single dose of 400 mg in adults or 12 mg/kg in children, on the first day of treatment.

Oropharyngeal Candidiasis

Adults: 100mg once daily. Treatment should be continued for at least 2 weeks to decrease the likelihood of relapse.
Children: 3 mg/kg once daily. Treatment should be continued for at least 2 weeks to decrease the likelihood of relapse.

Esophageal Candidiasis

Adults: 100 mg to 200 mg once daily. Patients should be treated for a minimum of 3 weeks and for at least 2 weeks following resolution of symptoms.
Children: 3 mg/kg to 6 mg/kg once daily. Patients should be treated for a minimum of 3 weeks and for at least 2 weeks following resolution of symptoms.

Systemic Candidiasis (Candidemia and Disseminated Candidal Infections)

Adult: 200 mg to 400 mg once daily. These patients should be treated for a minimum of 4 weeks and for at least 2 weeks following resolution of symptoms.
Children: 6 to 12mg/kg/day have been used in an open, noncomparative study of a small number of patients.

Cryptococcal Meningitis

Adults: 200 mg to 400 mg once daily. Although the duration of therapy for cryptococcal meningitis is unknown, it is recommended that the initial therapy should last a minimum of 10 weeks.
Children: 6 to 12 mg/kg once daily. The recommended duration for initial therapy is 10 to 12 weeks after the cerebrospinal fluid becomes culture-negative.

Prevention of Recurrence of Cryptococcal Meningitis in Patients with AIDS

Adults: 200 mg once daily.
Children: 6 mg/kg once daily.

Premature Neonates

Experience with fluconazole in neonates is limited to pharmacokinetic studies in premature newborns. Based upon the prolonged half-life seen in premature newborns (gestational age 26 to 29 weeks), these children in the first 2 weeks of life, should receive the same dosage (mg/kg) as in older children, but administered every 72 hours. After the first 2 weeks, these children should be dosed once daily.
Neonates: No information regarding fluconazole pharmacokinetics in full-term newborn is available.

Prophylaxis in Adult Patients

The recommended fluconazole daily dosage for the prevention of candidiasis in adult patients undergoing bone marrow transplantation is 400 mg once daily. Patients who are anticipated to have severe granulocytopenia (less than 500 neutrophils/mm³) should start fluconazole prophylaxis several days before the anticipated onset of neutropenia and continue for 7 days after the neutrophil count rises above 1000 cells/mm³.
Fluconazole may be administered either orally or by intravenous infusion. If intravenous infusion is considered, then the Product Monograph for that specific product should be consulted.

Dosage in Patients With Impaired Renal Function

Adults: Fluconazole is cleared primarily renal excretion as unchanged drug. In patients with impaired renal function, an initial loading dose of 50 to 400 mg should be given. After loading dose, the daily dose (according to indication) should be based on the following table:

Creatinine Clearance (mL/min)	Percent of Recommended Dose
> 50	100%
21 - 50	50%
11 – 20	25%
Patients receiving regular hemodialysis	One recommended dose after each dialysis

Children: Although the pharmacokinetics of fluconazole has not been studied in children with renal in sufficiency, dosage reduction in children with renal insufficiency should parallel that recommended for adults.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

In the event of overdose, symptomatic treatment (with supportive measures and gastric lavage if necessary) may be adequate. Fluconazole is largely excreted in urine. A three hour hemodialysis session decreases plasma levels by approximately 50%.

STORAGE CONDITIONS: Store below 30°C

AVAILABILITY OF DOSAGE FORMS
Apo-Fluconazole 50mg: Pink, trapezoid, biconvex tablet, engraved 'APO' over '50' on one side, other side plain. Available in HDPE bottles of 50's.
Apo-Fluconazole 100mg: Pink, trapezoid, biconvex tablet, engraved 'APO' over '100' on one side, other side plain. Available in HDPE bottles of 50's.

Manufacturer: Apotex Inc., 150 Signet Drive, Weston Ontario, Canada M9L 1T9.

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